

Detection of Plastic Litter on Beaches



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I. Introduction

Plastic pollution is one of the most critical environmental problems, threatening marine ecosystems, wildlife, and human health. According to Our World In Data, annual plastic production has escalated dramatically, experiencing an almost 230-fold increase from 2 million tonnes in the 1950s to 459 million tonnes in 2019. Specifically during 2019, more than 970 thousand tonnes of plastic waste entered the ocean. The presence of plastic waste on beaches and ocean is troubling due to a higher likelihood of ingestion by marine animals and birds.

Recent research has quantified the mortality risk of plastic ingestion across taxa. The study analyzed more than 10000 necropsies of seabirds, marine animals, and sea turtles. It showed that ingestion of macroplastic, such as hard and soft plastics, rubber and fishing debris leads to acute mortality by obstructing, perforating or twisting the gastrointestinal tract. The study also estimated that ingesting anywhere from 6 to 405 pieces of macroplastic of various sizes can result in a 90% probability of death. These findings strongly emphasized the urgent need of monitoring and mitigating plastic litter in coastal environments.

Detection of plastic litter on beaches offers a versatile and efficient way to measure and monitor plastic litter over the traditional manual surveys, which tends to be time-consuming, labor-intensive, and prone to human error. Consequently, this project utilizes the BePLi v1 dataset, to develop an object detection model to detect plastic litter in an effort to reduce wildlife exposure and environmental risks.

Research Objectives

The objectives of this study are as follows:

1. To develop a deep learning-based object detection model for identifying plastic litter in beach environments using the YOLOv8 architecture.
2. To analyze the performance of the proposed model in detecting plastic litter under complex visual conditions, including background clutter, scale variation, and partial occlusion.
3. To evaluate the model's detection performance using standard object detection metrics

II. Dataset

The dataset used in this project is the BePLi v1 dataset, which consists of 3709 images of beach plastic litter collected from beaches in Yamagata Prefecture, Japan. Each image is accompanied by manually annotated labels. The dataset contains only a single class, which is plastic litter. The dataset is organized into two directories: images and labels. The image directory is subdivided into original_images, train, test, and val folders, containing 3709, 2226, 742, and 741 images, respectively. The labels directory contains JSON files corresponding to the train, test, and val splits in a partially modified MS COCO format.

III. Modeling

This project uses a pre-trained model, which is the You Only Look Once (YOLO) model, specifically YOLOv8n as the main model. This model is chosen due to its balance between speed and accuracy, making it suitable for real-time applications. The architecture is divided into three main components, which is Backbone, Neck, and Head.

1. The Backbone utilized Convolutional Neural Network (CNN) to extract multi-scale features from the input image.
2. The Neck aggregates the extracted features using an optimized version of the Path Aggregation Network (PANet), improving the detection of objects at different sizes and scales.
3. The Head predicts the bounding box coordinates, confidence score, and object class based on the extracted features.

The model was fine-tuned on the BePLi v1 dataset using hyperparameters such as 200 training epochs, a batch size of 32, and an input image size of 640x640. Furthermore, the fine tuning process is illustrated in Figure 1.

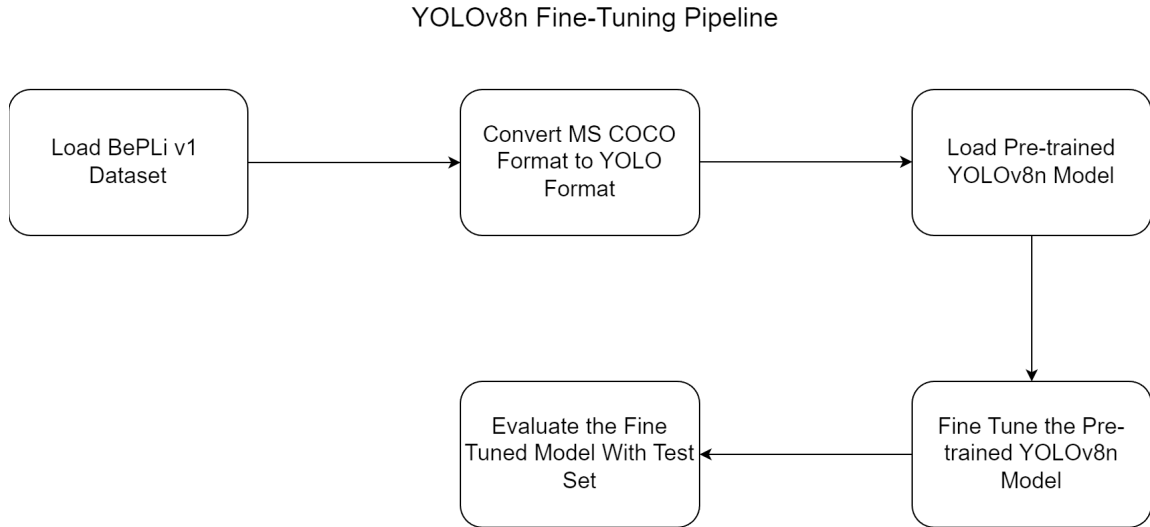


Figure 1. YOLOv8n Fine Tuning Pipeline

The fine tuning process started by loading the BePLi v1 dataset and converting the annotations from MS COCO format to YOLO format, which is required for training the YOLO model. Then, the pre-trained YOLOv8n model was loaded and fine tuned on the BePLi v1 train set. Lastly, the model was evaluated using the test set to obtain the evaluation results, including Precision, Recall, F1 Score, mAP50, and mAP50-95.

Model Architecture

To provide a clearer overview of the YOLOv8n architecture used in this project, the structural composition of the backbone, neck, and head components is illustrated. The architecture highlights the flow of feature extraction, multi-scale feature aggregation, and detection heads responsible for predicting bounding boxes at different resolutions.

As shown in Figure 2, the backbone extracts hierarchical spatial features, which are then aggregated by the neck through multi-scale feature fusion before being passed to the detection head.

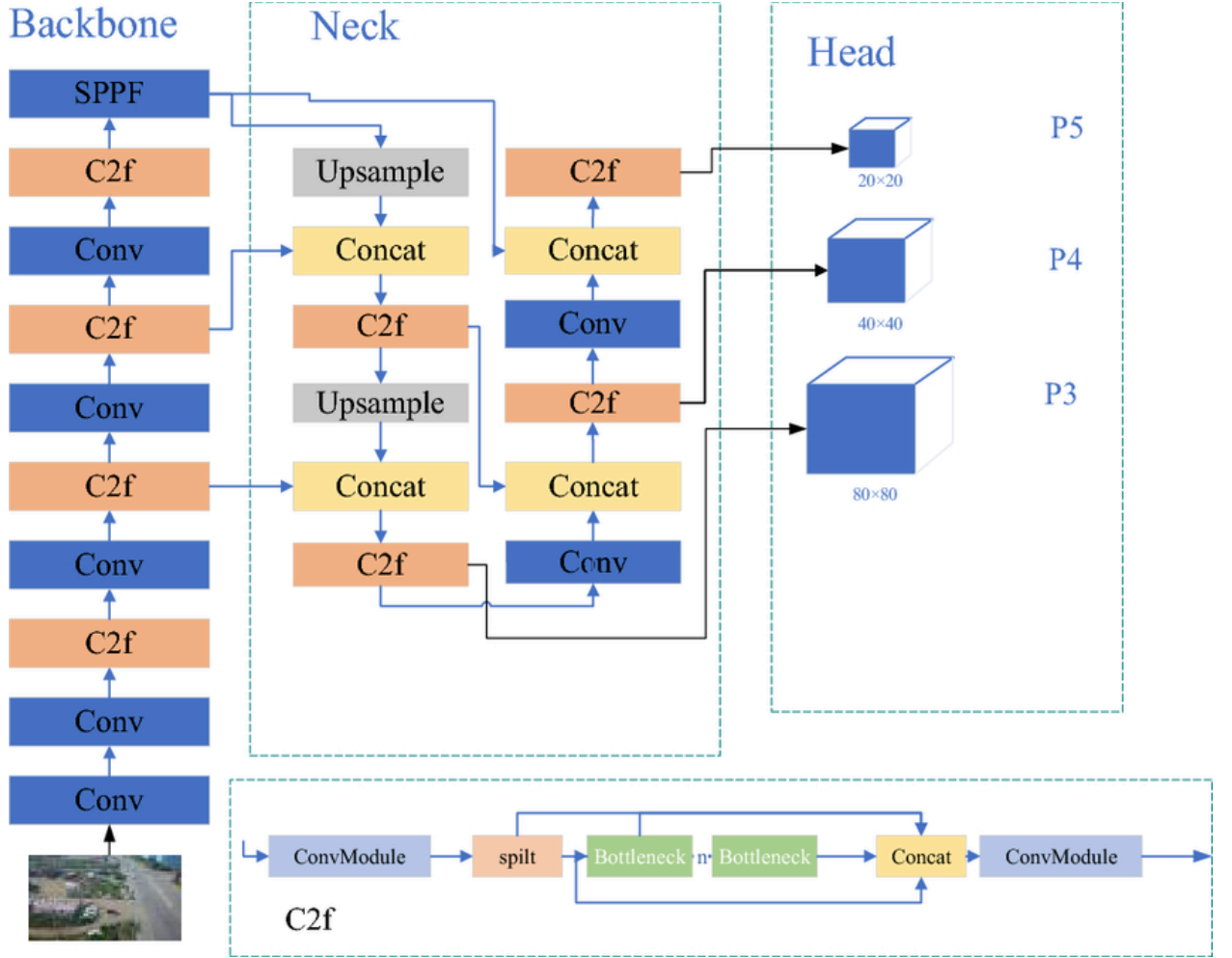


Figure 2. Architecture of the YOLOv8n model illustrating the backbone, neck, and head components used for beach plastic litter detection.

Backbone

The purpose of the backbone is to extract hierarchical spatial features from the imputed image. YOLOV8 backbone consist of convolution layers, batch normalization, SILU activation, and C2f (cross stage partial with feature fusion)

- Convolution layers

Let the input feature map be denoted as $X \in R^{H \times W \times C}$. Each convolutional layer applies a set of learnable filters to produce an output feature map:

$$Y_{i,j}^{(k)} = \sum_{m,n,c} X_{i+m, j+n, c} \times W_{m,n,c}^{(k)} + b^{(k)}$$

Where:

- $W^{(k)}$ denotes the convolutional kernel for the k -th output channel,
- $b^{(k)}$ is the bias term,
- (m, n) index the spatial extent of the kernel,
- c indexes the input channels.

This operation enables the backbone to extract spatial features of plastic litter that are invariant to translation and robust to background clutter commonly found in beach environments, such as sand, rocks, and organic debris.

- **Batch normalization**

To stabilize training and accelerate convergence, batch normalization is applied after the convolution operation. In convolutional neural networks, batch normalization is performed per feature channel and is defined as:

$$\hat{x}^{(k)} = \frac{x^{(k)} - \mu^{(k)}}{\sqrt{(\sigma^{(k)})^2 + \epsilon}}, y^{(k)} = \gamma^{(k)} \hat{x}^{(k)} + \beta^{(k)}$$

Where $\mu^{(k)}$ and $\sigma^{(k)}$ denote the batch mean and variance of the k -th feature channel, while $\gamma^{(k)}$ and $\beta^{(k)}$ are learnable scaling and shifting parameters.

Batch normalization reduces internal covariate shift, leading to more stable gradient propagation and faster convergence during training. This is particularly important when fine-tuning the model for 200 training epochs, as it allows the use of higher learning rates without causing training instability.

- **SILU Activation Function**

YOLOv8 employs the Sigmoid Linear Unit (SiLU) activation function, defined as:

$$SILU(x) = x \times \sigma(x) \text{ with } \sigma(x) = \frac{1}{1 + e^{-x}}$$

SiLU provides smoother gradients compared to ReLU-based activations, improving optimization stability when learning complex visual patterns of plastic litter.

- **C2f Feature Fusion Module**

To balance representational power and computational efficiency, YOLOv8 utilizes the C2f module, which enables partial feature reuse and progressive feature fusion. Let the intermediate feature transformations be denoted as:

$$f_1 = g_1(X), f_2 = g_2(f_1), \dots, f_n = g_n(f_{n-1})$$

The output of the C2f module is formed by concatenating selected intermediate feature maps:

$$F = \text{Concat}(f_1, f_2, f_3, \dots, f_n)$$

This design allows the backbone to preserve low-level spatial details while progressively incorporating higher-level semantic information, which is crucial for detecting plastic litter objects with varying shapes and scales.

Neck

The neck component of YOLOv8 serves as a critical bridge between feature extraction and object detection. While the backbone focuses on learning hierarchical representations, the neck is responsible for aggregating features from multiple spatial resolutions to ensure that both fine-grained details and high-level semantic information are preserved.

This function is particularly important for beach plastic litter detection, as plastic objects often appear in a wide range of sizes and are frequently partially occluded or visually similar to their surrounding background.

YOLOv8 adopts principles from the Feature Pyramid Network (FPN) and Path Aggregation Network (PAN) to propagate information across different feature scales in both top-down and bottom-up directions.

- **Upsampling**

Let $X \in R^{H \times W \times C}$ represent a low-resolution feature map extracted from deeper layers of the backbone. To match the spatial resolution of higher-resolution features, an upsampling operation is applied:

$$X_{up} = Upsample(X, s)$$

where s denotes the upsampling scale factor.

This step increases the spatial resolution of semantically rich features, allowing small plastic objects that may be lost in deeper layers to remain detectable.

- **Feature Fusion**

After upsampling, feature maps from different levels are combined using concatenation:

$$F_{fused} = Concat(F_{low}, F_{high})$$

Here, F_{low} contains fine spatial details, while F_{high} provides stronger semantic context. Concatenation preserves information from both feature sources, enabling better detection in visually cluttered beach environments.

- **Convolution Refinement**

The fused feature map is refined through convolutional layers:

$$F_{out} = Conv(F_{fused})$$

This refinement helps the model emphasize relevant object features while suppressing background noise. The resulting multi-scale feature maps are then passed to the detection head for bounding box prediction.

Head

The head component of YOLOv8 is responsible for producing the final detection outputs, including bounding box coordinates and object confidence scores. Unlike the backbone and neck, which focus on feature extraction and aggregation, the head directly performs object detection.

In this project, the BePLi v1 dataset contains only a single class (plastic litter). Therefore, the detection head primarily focuses on accurate localization and object presence estimation rather than multi-class classification.

- **Bounding Box Regression**

For each spatial location on the feature map, YOLOv8 predicts a bounding box parameterized as:

$$\hat{b} = (x, y, w, h)$$

Where:

- (x, y) represent the center coordinates of the bounding box,
- w and h represent the width and height of the bounding box, normalized with respect to the input image size.

These parameters are regressed directly from the feature maps produced by the neck, enabling YOLOv8 to localize plastic litter objects in a single forward pass.

- **Objectness Confidence**

In addition to bounding box coordinates, the model predicts an objectness score that represents the probability of an object being present within the predicted bounding box. This score is computed using a sigmoid activation function:

$$P_{obj} = \sigma(s)$$

Where s is the raw output score from the detection head.

Because the dataset is single-class, the objectness score plays a central role in determining whether a detected region corresponds to plastic litter.

- **Classification Loss**

Since the BePLi v1 dataset contains only one object class, the class probability is defined as:

$$P_{class} = 1$$

As a result, the classification branch contributes minimally to the detection process, and the overall prediction confidence is largely governed by the objectness score and bounding box quality.

- Intersection over Union (IoU)

The accuracy of bounding box localization is measured using Intersection over Union (IoU), defined as:

$$IoU = \frac{Area(B_{pred} \cap B_{gt})}{Area(B_{pred} \cup B_{gt})}$$

IoU quantifies how well the predicted bounding box B_{pred} overlaps with the ground truth bounding box B_{gt} .

- Loss Function

The overall loss function used to train the detection head is expressed as:

$$L = L_{box} + L_{obj} + L_{cls}$$

Where:

- L_{box} represents the bounding box regression loss,
- L_{obj} represents the objectness loss,
- L_{cls} represents the classification loss.

Given the single-class nature of the dataset, the classification loss L_{cls} has minimal impact on the total loss, while the bounding box regression loss L_{box} becomes the dominant factor influencing model performance.

$$L_{box} = 1 - IoU$$

This formulation explains why the model achieves moderate detection performance while struggling with precise localization, particularly for small or partially occluded plastic litter.

The relatively low mAP_{50-95} score indicates that although the model can detect the presence of plastic litter, the predicted bounding boxes often do not tightly align with the ground truth annotations. This suggests that localization error, rather than object presence detection, is the primary limitation of the model.

IV. Evaluation

The fine-tuned YOLOv8n model was assessed using Precision, Recall, F1 Score, mAP50, and mAP50-95. Precision measures the proportion of the correctly predicted positives among all positive predictions, showing how accurately the model predicts an object. Recall measures the proportion of correctly predicted positives among all actual positives, indicating how well the model detects all positive objects. Meanwhile, F1 Score combines Precision and Recall to measure how the model is able to trade off between precision and recall in a balanced manner. This is because an increase in precision would reduce recall, and vice versa. Furthermore, mAP50 is the mean Average Precision at a single Intersection over Union (IoU) threshold of 0.5, where a detection is considered correct if its IoU is greater than or equal to 0.5. For context, Average Precision measures the area under the Precision Recall (PR) Curve, while Intersection over Union (IoU) measures the overlap between the predicted bounding box and the ground truth bounding box. Lastly, mAP50-95 is the mean of Average Precision calculated at multiple IoU thresholds between 0.5 and 0.95 in increments of 0.05.

The quantitative performance of the fine-tuned YOLOv8n model was evaluated using standard object detection metrics, as summarized in Table 1.

Table 1. Quantitative evaluation results of the fine-tuned YOLOv8n model for beach plastic litter detection.

Metric	Description	Value	Interpretation
Precision	Proportion of correctly predicted positives among all positive predictions.	0.52	Approximately half of the positive predictions are correct.
Recall	Proportion of correctly predicted positives among all actual positives.	0.40	Only 40% of the actual plastic litter instances are effectively captured.

F1 Score	Combines Precision and Recall to measure the balanced trade-off.	0.45	Reflects a moderate overall performance.
mAP50	Mean Average Precision at IoU threshold of 0.5.	0.39	The model correctly detects 39% of the litter instances with sufficient overlap, suggesting it detects many objects but struggles with localization.
mAP50-95	Mean of Average Precision calculated at multiple IoU thresholds (0.5 to 0.95).	0.17	Indicates that many predicted bounding boxes did not precisely align with the ground truth bounding box.

The fine-tuned YOLOv8n model evaluation results shows that the model attains Precision of 0.52 and Recall of 0.40. This suggests that approximately half of the positive predictions are correct, whereas only 40% of the actual plastic litter instances are effectively captured. The F1 Score of 0.45 reflects a moderate overall performance, underscoring an inherent trade off between Precision and Recall. Furthermore, the model obtained an mAP50 of 0.39, indicating that the model correctly detects 39% of the litter instances with sufficient overlap. This shows that the model is able to detect many objects, but struggle with localization. Lastly, the mAP50-95 of 0.17 indicates that many predicted bounding boxes didn't precisely aligned with the ground truth bounding box. In conclusion, the model is able to detect the plastic litter moderately but struggles with precise bounding box prediction on smaller or occluded plastic litter.

V. Deployment

The fine-tuned model was deployed as a website application using Streamlit. The application allows the user to upload an image of a beach containing plastic litters, and the model will perform an inference and output the predicted bounding box and the corresponding confidence score as illustrated in Figure 3.

Beach Plastic Litter Detection Workflow (Streamlit)



Figure 3. Workflow of Beach Plastic Litter Detection Web Application Using Streamlit

VI. Reflection

This project demonstrates the practicality of utilizing deep learning-based object detection to detect plastic litter on beaches by using BePLi v1 dataset and YOLOv8n model. The model was able to detect plastic litter with moderate accuracy while still being lightweight, making it ideal for real-world deployment. Furthermore, the deployment into a web application using Streamlit highlights the feasibility of employing Artificial Intelligence for environmental monitoring. However, we experienced several challenges in this project.

The evaluation results indicate that the model performed moderately, resulting in low Precision, Recall, F1 Score, mAP50, and mAP50-95, in which the model struggles to detect all plastic litter and localize the bounding box precisely. These challenges could be tackled by expanding the dataset to incorporate more diverse beach conditions, lighting conditions, and plastic litter classes. Furthermore, utilizing larger YOLO variant models may further improve the accuracy. Overall, this project highlights both the potential and the challenges of applying a deep learning-based object detection model to real world environmental problems.

VII. Related Links

The source code for the project is hosted in Github and can be accessed at <https://github.com/Alvindral/Deep-Learning-Beach-Waste-Object-Detection-Using-YOLOv8n>.

VIII. References

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